「2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p-subgroups.
By the Sylow theorem, $n_{101} \equiv 1 \bmod 101$ and $n_{101} \mid 20$, so n_{101} must be 1.
Therefore, the Sylow 101-subgroup is normal, so proved.

「Without using Sylow theorem.

By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.
Denote $A = \{gHg^{-1} \mid g \in G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \to \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\# A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \# A \leq 20.$$

Now, consider the action

$$H \times A \to A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H-orbit of this action is $\mathrm{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\# \mathrm{Orb}_H(K) = |H : \mathrm{Stab}_H(K)|, \ \mathrm{Stab}_H(K) \leq H \Rightarrow |\mathrm{Stab}_H(K)| \in \{1, 101\}.$$

So, $\# \mathrm{Orb}_H(K) \in \{1, 101\}$ and $\mathrm{Orb}_H(K) \leq A$, so $\# \mathrm{Orb}_H(K) = 1$.
therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.
Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{H\}$. That is, $H \trianglelefteq G$. ☒

[2021. spring

17. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \cdots + a_d z^d \quad \text{over } \mathbb{C}, \ a_d \neq 0.$$

$\boxed{8m.}$

Evaluate the value $\displaystyle\lim_{R\to\infty}\int_{|z|=R} \frac{z\cdot f'(z)}{f(z)}\,dz$

Proof. By the fundamental theorem of Algebra, f has d-roots $z_1, \cdots, z_d \in \mathbb{C}$,

That is, $f(z) = a_d(z-z_1)\cdots(z-z_d)$

Moreover, $\dfrac{f'(z)}{f(z)} = \dfrac{1}{z-z_1} + \cdots + \dfrac{1}{z-z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \cdots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i}\int_{|z|=R} \frac{z\cdot f'(z)}{f(z)}\,dz = \sum_{n=1}^{d} \operatorname{Res}\left(\frac{z}{z-z_n}, z_n\right) = \sum_{n=1}^{d} \lim_{z\to z_n} z = z_1 + \cdots + z_n$$

Consequently, the value is

$$2\pi i\cdot(z_1 + \cdots + z_n) = 2\pi i\cdot\left(\frac{-a_{d-1}}{a_d}\right),$$

$\boxtimes$

[2022. Spring.

8. Define $f : \mathbb{R} \to \mathbb{R} ; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$

Then, f is differentiable on $\mathbb{R}$.

proof. For each $m \in \mathbb{N}$, define

$$f_m(x) = \sum_{n=1}^{m} \frac{1}{x^2+n^2}.$$

At first, $f_m(0) \to f(0)$ as $m \to \infty$, because that

$$\sum_{n=2}^{m} \frac{1}{n^2} < \sum_{n=2}^{m} \frac{1}{n^2-n} = \sum_{n=2}^{m} \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \to 1 \text{ as } m \to \infty,$$

so the increasing bounded sequence $f_m(0)$ converges. Now,

$$f_m'(x) = \sum_{n=1}^{m} - \frac{2x}{(x^2+n^2)^2}$$

converges uniformly, because

if $|x| \le 1$, then

$$\left| \frac{-2x}{(x^2+n^2)^2} \right| \le \frac{2}{n^4} \quad \text{and} \quad \sum \frac{1}{n^4} < \infty,$$

if $|x| > 1$, then

$$\left| \frac{-2x}{(x^2+n^2)^2} \right| \le 2 \cdot \left| \frac{x}{x^2+n^2} \cdot \frac{1}{x^2+n^2} \right| \le 2 \cdot \frac{1}{n^2} \quad \text{and} \quad \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives f_m' converges uniformly.

Now, f_m converges at some point and f_m' converges uniformly on $\mathbb{R}$, the limit function f is differentiable.] gm.

First, for any $x \in \mathbb{R}$, $|x^2+n^2| \ge n^2$, so

$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2+n^2} \right| \le \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^{m} \frac{1}{n^2} < \sum_{n=2}^{m} \frac{1}{n^2-n} = \sum_{n=2}^{m} \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \to 1,$$

as $m \to \infty$. Therefore, given $\varepsilon > 0$, for some $N \in \mathbb{N}$, $m > n \ge N$ implies

$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^{m} \frac{1}{x^2+k^2} \right| \le \sup_{x \in \mathbb{R}} \sum_{k=n}^{m} \left| \frac{1}{x^2+k^2} \right| < \sum_{k=n}^{m} \frac{1}{k^2} < \varepsilon.$$

Now, define for each $m \in \mathbb{N}$,

$$f_m(x) = \sum_{n=1}^{m} \frac{1}{x^2+n^2}.$$

We already shown $f_m \to f$ uniformly. And,

$$f_m'(x) = \sum_{n=1}^{m} - \frac{2x}{(x^2+n^2)^2}$$

Converges uniformly, because

2022. Fall.

8. For Continuous functions $g:[0,1]\to\mathbb{R}$ and $h:[-1,1]\to\mathbb{R}$, define

$$f:[0,1]\to\mathbb{R}: x\mapsto\int_0^1 h(x-y)g(y)\,dy$$

For each $n\in\mathbb{N}$, define

$$f_n:[0,1]\to\mathbb{R}: x\mapsto\frac{1}{n}\cdot\sum_{i=0}^{n-1}h\left(x-\frac{i}{n}\right)\cdot g\left(\frac{i}{n}\right)$$

Prove that $f_n\to f$ uniformly on $[0,1]$

Proof. Define $H:[0,1]\times[0,1]:(x,y)\mapsto h(x-y)\cdot g(y)$.

Since h and g both are continuous, therefore H is continuous and Since $[0,1]^2$ is compact set, so H is continuous uniformly.

Given $\varepsilon>0$, there exists $\delta>0$ such that for any $(x_1,y_1),(x_2,y_2)\in[0,1]^2$,

$$\|(x_1,y_1)-(x_2,y_2)\|<\delta\Rightarrow\|H(x_1,y_1)-H(x_2,y_2)\|<\varepsilon.$$

Now, given $x\in[0,1]$, and $n>\frac{1}{\delta}$, ($n>\frac{1}{\delta}$ gives $\frac{1}{n}<\delta$)

$$\begin{aligned}|f(x)-f_n(x)|&=\left|\int_0^1 H(x,y)\,dy-f_n(x)\right|\\&=\left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}H(x,y)\,dy-\frac{1}{n}\cdot\sum_{i=0}^{n-1}H\left(x,\frac{i}{n}\right)\right|\\&=\left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}H(x,y)\,dy-\sum_{i=1}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}H\left(x,\frac{i}{n}\right)dy\right|\\&\le\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}\left|H(x,y)-H\left(x,\frac{i}{n}\right)\right|dy\\&<\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}\varepsilon\,dy=\varepsilon.\end{aligned}$$

Therefore, $n>\frac{1}{\delta}\Rightarrow\sup_{x\in[0,1]}|f_n(x)-f(x)|<\varepsilon$, So, proved. ∎

⌈2022. Fall.

13. Let S be the subset of $\mathbb{R}^3$ given by $\{(x,y,z) \mid z^2 = x^2+y^2,\ z \geq 0\}$.
Show that S is not regular surface.

「2024, Spring.

1. Let $A, B \in M_{m\times n}(F)$. Show that $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$. $\boxed{5m}$

Proof. By definition,

$$\operatorname{rank}(A+B) = \dim \operatorname{Im}(A+B) \text{ and } \operatorname{rank} A = \dim \operatorname{Im} A, \ \operatorname{rank} B = \dim \operatorname{Im} B.$$

Given $\alpha \in \operatorname{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \operatorname{Im} A + \operatorname{Im} B$ for some $\beta \in F^m$, therefore $\operatorname{Im}(A+B) \subseteq \operatorname{Im} A + \operatorname{Im} B$. Now,

$$\operatorname{rank}(A+B) \le \dim(\operatorname{Im} A + \operatorname{Im} B) \le \dim \operatorname{Im} A + \dim \operatorname{Im} B = \operatorname{rank} A + \operatorname{rank} B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ such that $P(\alpha_i) = c_i\alpha_i$ for some $c_i > 0$, for each i. $\boxed{15m}$

Let $1 \le i \le n$ be fixed. Then, by the assumption,

$$P^2\alpha_i = c_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - c_i^2 I)\alpha_i = (Q - c_i I)(Q + c_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - c_i I)\alpha_i = 0$, because; if it is not zero, then

$$(Q - c_i I)(Q + c_i I)\alpha_i = (Q + c_i I)(Q - c_i I)\alpha_i = 0,$$

So $-c_i$ is negative eigenvalue of Q, it contradicts to Q is positive definite. Therefore, for each $1 \le i \le n$,

$$P\alpha_i = c_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. ∎

「2024, Spring

3. Let $T: \mathbb{R}^5 \to \mathbb{R}^5 ; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T-invariant subspace $V \subseteq \mathbb{R}^5$ s.t $\dim V = 4$?

b). 〃 s.t $\dim V = 2$, $T|_V$ is diagonalizable?

c) 〃 $\dim V = 3$, 〃

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, \ Ae_2 = e_1 + e_2, \ Ae_3 = e_3, \ Ae_4 = e_3 + e_4$$

are all contained in V, so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V, and

$$[T|_V]_{\{e_1, e_3\}} = I_2.$$ So $T|_V$ is diagonalizable.

c). Observe characteristic polynomial of T is $(t-1)^5$.
If such a subspace V exists, then the characterstic polynomial of $T|_V$ is $(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$.
That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A - I) = \dim \ker \begin{pmatrix} 0 & 1 & & & \\ & 0 & 0 & & \\ & & 0 & 1 & \\ & & & 0 & 1 \\ & & & & 0 \end{pmatrix} = 2.$$

「2024 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$.
For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f,g) := \sup_{x\in\mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $m,n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n\to\infty} d(f, f_n) = 0$

b). Prove that $\lim_{n\to\infty}\left(\sup_{x\in\mathbb{R}} |f_n(x)|\right) = \sup_{x\in\mathbb{R}} |f(x)|$.

Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis,
Since $\mathbb{R}$ is Complete, $\lim_{n\to\infty} f_n(x)$ exists, Put this $f(x) \in \mathbb{R}$.

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \to 0$. Enough to show that

$$\sup_{x\in\mathbb{R}} |f(x) - f_n(x)| \to 0 \quad \text{as } n\to\infty.$$

「2025. Fall

1. a). Show that $\phi : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p'(0)$ is linear functional. $\boxed{8m}$

b). Find a dual basis corresponding to $\{1, t, \cdots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{n} b_i x^i$ and $c \in \mathbb{R}$ be given.

Then,
$$\phi(cf+g) = (cf+g)'|_{t=0} = c \cdot f'|_{t=0} + g'|_{t=0} = c \cdot \phi(f) + \phi(g)$$
by the linearity of differentiation. This shows the a).

To show b), let
$$\beta^* := \left\{ \frac{1}{k!} \cdot \phi_k : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p^{(k)}(0) \;\middle|\; k = 0, 1, \cdots, n \right\}$$
Then, for each $k = 0, 1, \cdots, n$,
$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$
Similarly a), the elements in β^* are all linear functional, so the β^* is dual basis.

「2025. Fall

2. Let $A \in M_{m\times m}(\mathbb{R})$, $\boxed{15m}$: $A^{\underline{n}}$ 이라 얼큰, $n=2$인때부터 생각

a). For $n \in \mathbb{N}$, $\operatorname{rank} A^n \leq \operatorname{rank} A$.

b). Let $L_A : \mathbb{R}^m \to \mathbb{R}^m ; x \mapsto Ax$.

Show that for $n \geq 2$,

$$\operatorname{rank} A^n = \operatorname{rank} A \text{ implies } \operatorname{Ker} L_A \cap \operatorname{Im} L_A = \{0\}.$$

Sol) a). Given $x \in \operatorname{Im} A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.
So, $x = A^n y = A^{n-1} Ay \in \operatorname{Im} A$, for $n \geq 2$. Thus $\operatorname{Im} A^n \subseteq \operatorname{Im} A$, it follows
$\operatorname{rank} A^n \leq \operatorname{rank} A$. If $n=1$, Clear.

b). Given $x \in \operatorname{Ker} A$, $A^n x = A^{n-1} Ax = A^{n-1} 0 = 0$, so $x \in \operatorname{Ker} A^n$, so $\operatorname{Ker} A \subseteq \operatorname{Ker} A^n$
Since $\operatorname{rank} A^n = \operatorname{rank} A$, the dimension theorem gives $\operatorname{nullity} A^n = \operatorname{nullity} A$.
Therefore, since $\mathbb{R}^m$ is finite dimensional, So is $\operatorname{Ker} A^n$, thus $\operatorname{Ker} A = \operatorname{Ker} A^n$.
Now, let $x \in \operatorname{Ker} L_A \cap \operatorname{Im} L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.
Then, $A^{n-1} \cdot x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} \cdot Ax = 0$, and this implies $y \in \operatorname{Ker} A^n$.
Since $\operatorname{Ker} A^n = \operatorname{Ker} A$, $Ay = 0$, so $x = Ay = 0$, so proved. ⊣

「2025. Fall

3. Find all $a,b,c \in \mathbb{R}$ such that $\boxed{6m}$

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix}$$ is diagonalizable.

Sol). Since the charaterstic polynomial of A is regardless of a, b, c,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the charaterstic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

Calculating gives

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a,b,c).

「2025. Fall」

20m

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that

$$\det A = \pm \det B.$$

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

and $\det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$

so, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,
first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.
therefore, by the subgroup criterion, N is subgroup, and normal. 10m.
Define a map

$$\varphi : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to \mathbb{R}^\times : (A, B) \mapsto (\det A \cdot \det B^{-1})^2$$

Then, $\ker \varphi = N$ and φ is homomorphism, onto $(0, \infty)$, because the det preserves multiplication, so is φ, and ontoness is given by: fix $B = I$, and $A = \mathrm{diag}(\sqrt{r}, 1, 1, \cdots, 1)$ where $r > 0$ is arbitrary.
By the first-isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times)$$ 10m. φ

Comment: φ 을 먼저 찾고 시작하고, hom 임을 보이면 앞 계산 과정이 날라감!

[2025. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G.

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.
By the Cauchy's theorem, G has order 3 element x and order 7 element y.
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \mod 21 \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g, $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \mod n$ has a solution iff $\gcd(a, n) \mid b$.
And in such a case, it has $\gcd(a, n)$ solution

「2025. Fall.

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

$\boxed{22m}$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n\to\infty} a_n = \sup\{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0,1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.
Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.
Let a be a limit point of $\{a_n\}^*$. Then, given $k \in \mathbb{N}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
For $k=1$, choose such a $p_1 \in B_1(a) \setminus \{a\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{1-|p-a|}(p)$, then $a_{p_1} \in B_1(a)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
Then, similarly, there is $a_{p_k} \in B_{\frac{1}{k}}(a)$ such that $p_k > p_{k-1}$,

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point a is contained in $\{a_n\}^*$,
So it is closed. ◻

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup\{a_n\}^* = \max\{a_n\}^* = a \text{ for some subsequential limit } a.$$

So, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < a + \varepsilon$ (Since $a = \max\{a_n\}^*$)
and there exists infinitely many terms in $\{a_n\}$ such that $a - \varepsilon < a_n$ (Since $a_{n_k} \to a$)
Thus $\limsup_{n\to\infty} a_n = a$.

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \cdots$$

Then, given $b \in [0,1]$, there exists a subsequence in $\{b_n\}$ converges to b, because: for any $\varepsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \varepsilon,$$

So we can construct the subsequence. ◻

「2025, Fall,

17. Let E be a non empty bounded subset of a Metric space (X,d). 10m

Define $\quad \operatorname{diam} E := \sup\{d(x,y) \mid x,y \in E\}$

a). Prove $\operatorname{diam} E = \operatorname{diam} \overline{E}$

b). If E is compact, then $d(x',y') = \operatorname{diam} E$ for some $x', y' \in E$.

Proof.

a). Since $E \subseteq \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$ is clear.

To show the inverse inequality, using the property of supremum,

Given $\varepsilon > 0$, there exists $x,y \in \overline{E}$ such that

$$\operatorname{diam} \overline{E} - \frac{\varepsilon}{2} < d(x,y) \leq \operatorname{diam} \overline{E}$$

And, since $\overline{E}$ is closure, there are $x', y' \in E$ such that

$$d(x',x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y,y') < \frac{\varepsilon}{4}.$$

Now, $\operatorname{diam} \overline{E} < d(x,y) + \frac{\varepsilon}{2} < d(x,x') + d(x',y') + d(y',y) + \frac{\varepsilon}{2} < d(x',y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$, so proved.] 7m.

b), Since E is compact set, so $E \times E$ is also compact set.

Since the metric function d is continuous, it preserves compactness,

so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\operatorname{diam} E = \max d[E \times E] \in d[E \times E],$$

So there is $(x',y') \in E \times E$ such that $d(x',y') = \max d[E \times E]$.] 3m.

「2025. Fall.

8. Let $a>1$. Evaluate $\boxed{5m}$

$$\lim_{t\to 0^+}\int_t^{at}\frac{\cos x}{x}\,dx$$

proof. Consider the taylor expansion

$$\cos x=\sum_{n=0}^{\infty}(-1)^n\cdot\frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x}=\frac{1}{x}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots$

Now, $\int_t^{at}\frac{\cos x}{x}dx=\int_t^{at}\frac{1}{x}+\int_t^{at}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots\ dx$

$$=\ln a+\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}$$

(It is possible, because $\sum(-1)^n\cdot x^{2n-1}/(2n)!$ converges for all $x\in(\mathbb{R})$

Now, $\lim_{t\to 0^+}\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}dx=0$, so

the given limit is $\ln a$.

Revised in next.

「Observation for $\lim_{t\to 0^+}\int_t^{at}\frac{\cos x}{x}\,dx$.

As we know, the power series representation

$$\frac{1}{x}\cdot\cos x=\frac{1}{x}\cdot\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\frac{x^{2n-1}}{(2n)!}$$

is valid for all $x\neq 0$. Therefore, for ant $t>0$,

$$\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

$$=\int_t^{at}\frac{1}{x}\,dx+\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

Meanwhile, the series converges uniformly in $[t,at]$ for small enough $t>0$, specifically, $t<\frac{1}{a}$, then for each $n\in\mathbb{N}$,

$$\left|(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\right|\le\left|\frac{1}{(2n)!}\right| \quad \text{and} \quad \sum_{n=1}^{\infty}\frac{1}{(2n)!}<\infty,$$

$\frac{1}{2n}x^{2n}$

therefore the Weierstrass M-test gives the condition.

Thus, we can exchange the integral to the sum,

$$\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}(-1)^n\cdot\left(\frac{1}{2n\cdot(2n)!}\cdot\left(a^{2n}t^{2n}-t^{2n}\right)\right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\lim_{t\to 0^+}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{1}{2n\cdot(2n)!}\cdot t^{2n}(a^{2n}-1)$$

$$=\sum_{n=1}^{\infty}\lim_{t\to 0^+} \quad //$$

$$=\sum_{n=1}^{\infty}0$$

$$=0$$

Consequently, the given integral is

$$\int_t^{at}\frac{1}{x}\,dx=\ln at-\ln t=\ln a.$$

Note: $\frac{\cos x}{x}=\frac{1}{x}+\frac{\cos x-1}{x}$, $\left|\int_t^{at}\frac{\cos x-1}{x}\,dx\right|\le\int_t^{at}\frac{x}{2}\to 0$ as $t\to 0$.

KKG, 73p.

2.7.14.

If $\{x_n\} \subset \mathbb{R}$ converges to $x \in \mathbb{R}$, then

$$\bigcap_{k=1}^{\infty} \overline{\{x_n \mid n \geq k\}} = \{x\}.$$

TKKG,

2.7.7, and 2.7.20.

Let $A, B \subseteq \mathbb{R}^n$.

1) If A is open, then $A+B$ is open.

2). If A, B

TSNU. Qual.

Prove that $\lim_{b\to\infty} \int_0^b \frac{\sin x}{x}\,dx$ exists.

Proof. First, let $a_n = n\pi$ $(n \in \mathbb{N})$. Then, a sequence $\left\{ \int_0^{a_n} \frac{\sin x}{x}\,dx \right\}$ converges,
because: $\left| \int_{n\pi}^{(n+1)\cdot\pi} \frac{\sin x}{x}\,dx \right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x}\,dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi} \frac{\sin x}{x}\,dx = \sum_{k=1}^{n} \int_{a_{k-1}}^{a_k} \frac{\sin x}{x}\,dx$ is alternating series, so converges.

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4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T : G \to S_{G/H_1 \times G/H_2} : x \mapsto T_x \quad \text{where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\operatorname{Ker} T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

a). Given $g_1, g_2 \in G$, $T(g_1 g_2)(aH_1, bH_2) = (g_1 g_2 aH_1, g_1 g_2 bH_2)$

$$= T(g_1) \circ T(g_2)(aH_1, bH_2)$$

for each $a, b \in G$, so it preserves the operation.

And $g \in \operatorname{Ker} T \iff T_g = id \iff$ for all $a, b \in G$, $T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$

$\iff$ for all $a, b \in G$, $a^{-1} g a \in H_1$ and $b^{-1} g b \in H_2$

$\Rightarrow g \in H_1 \cap H_2$ (take $a = b = 1_G$)

b). By the first-isomorphism theorem,

$$G/\operatorname{Ker} T \cong \operatorname{Im} T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

Now, $|G : H_1 \cap H_2| \leq |G : \operatorname{Ker} T| \leq \# S_{G/H_1 \times G/H_2}$

because $\operatorname{Ker} T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G:A|$ is finite, then

$$G/A \to G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \subseteq B$,

$g_1 A = g_2 A \iff g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \iff g_1 B = g_2 B$

and surjectiveness is clear.

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8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x<0 \\ f(x) & \text{if } 0 \le x \le 1 \\ f(1) & \text{if } x>1 \end{cases}$$

For each $n \ge 1$, define

$$\rho_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} \rho_n(x-y) \tilde{f}(y)\, dy, \quad x \in \mathbb{R}.$

a). Show that $T_n f \to f$ on $[0,1]$,

b). Determine $T_n f \to f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \iff -\frac{1}{n}+x \le y \le \frac{1}{n}+x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y)\, dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\left| T_n f(x) - f(x) \right| = \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y)\, dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot f(x)\, dy \right|$$

$$= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot \left[\tilde{f}(y) - f(x) \right] dy \right|$$

$$\le \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)|\, dy$$

$$< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon$$

Therefore, $\sup_{[0,1]} |T_n f - f| \to 0$ as $n \to \infty$, so converges uniformly.